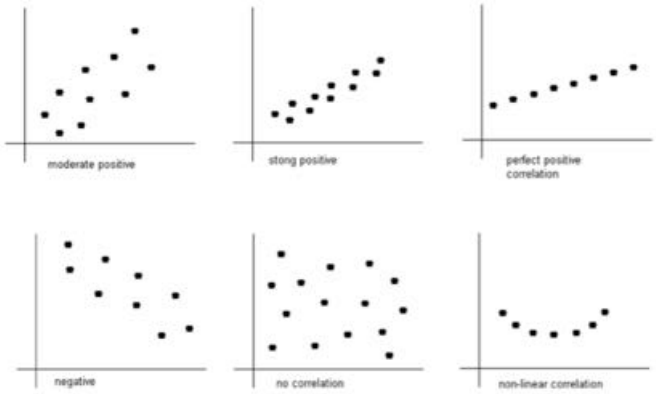
Correlation and Regression

Pearson Correlation **r**

https://en.wikipedia.org/wiki/Pearson_correlation_coefficient

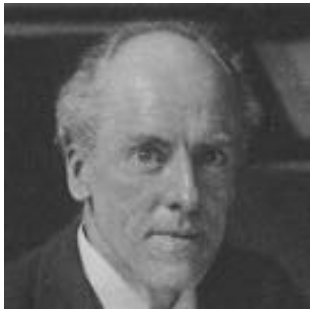
$$r = \frac{SP}{\sqrt{SS_X SS_Y}} = \frac{\sum(x-\bar{x})(y-\bar{y})}{\sqrt{\sum(x-\bar{x})^2 \sum(y-\bar{y})^2}}$$



$$\rho_{X,Y} = \frac{cov(X,Y)}{\sigma_X \sigma_Y} = \frac{E[(X-\mu_X)(Y-\mu_Y)]}{\sigma_X \sigma_Y}$$

Pearson correlation is also known under names:

- Pearson correlation coefficient (PCC)
- Pearson's r
- Pearson product-moment correlation coefficient (PPMCC)
- Bivariate correlation



Karl Pearson
(1857 - 1936)
Famous English mathematician and biostatistician

r can vary from -1 to 1:
-1 – perfect negative correlation
+1 – perfect positive correlation

R^2 is the square of the Pearson's correlation coefficient **r** (for a single regressor, fitted by least squares)

Linear Regression

Linear Regression = draw a line through experimental points.

- https://en.wikipedia.org/wiki/Linear_regression
- https://en.wikipedia.org/wiki/Simple_linear_regression
- https://paternogbc.shinyapps.io/SS_regression/ - play with it

We derive linear model by minimization the mean square error (MSE):

$$\min_{\hat{\beta}_0,\hat{\beta}_1} \sum_{i=1}^{N} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2.$$

We take partial derivatives by two coefficients - and solve system of 2 equations, and after some algebraic transformations, we derive the standard **OLS** method (**OLS = Ordinary Least Squares**).

y_m = y_mean = sum(y)/N
x_m = x_mean = sum(x)/N

var(x) = variance(x) = sum($(x-x_m)^2$)
covar(x,y) = covariance(x,y) = sum($(y-y_m)*(x-x_m)$)

slope b_1 = covar(x,y) / var(x)
intercept $b_0 = y_m - b_1 * x_m$

$$b_1 = \frac{\sum (x-\bar{x})(y-\bar{y})}{\sum (x-\bar{x})^2}$$

Standard Error of regression slope:

$$s(b_1) = \sqrt{\frac{1}{n-2} \cdot \frac{\sum (y_i - \hat{y}_i)^2}{\sum (x_i - \bar{x})^2}} \quad = \quad \sigma_{b_1} = \frac{\sigma_\varepsilon}{\sqrt{\sum (x-\bar{x})^2}}$$

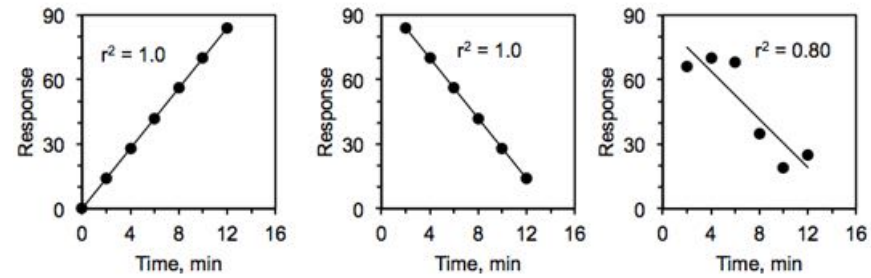
where sd(...) is standard deviation, and r – correlation coefficient

When drawing a curve through experimental points, it is very important to estimate how good this curve fits the data, and if fitting the data with this curve makes sense. For this you can use the **Coefficient of Determination R^2**:

– https://en.wikipedia.org/wiki/Coefficient_of_determination

"**R squared**" is a number between 0 and 1.
R^2 close to 1 – good model.

R^2 = **r^2** (the square of the Pearson's correlation coefficient **r** (for a single regressor, fitted by least squares)



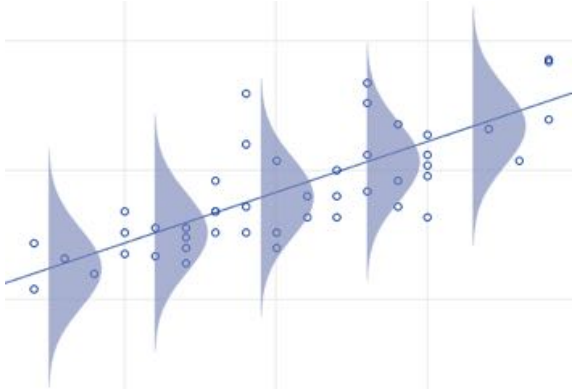
$$r^2 = \frac{\text{SSM}}{\text{SST}} = \frac{\sum (\hat{y}_i - \bar{y})^2}{\sum (y_i - \bar{y})^2}$$

$$R^2 = 1 - \frac{SS_{RES}}{SS_{TOT}} = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}$$

where:
SS = Sum of Squared
RES – residual errors = diffs between model and points)
TOT – total, diffs between mean and points
y_bar – mean
y_hat – model

Linear Regression - assumptions

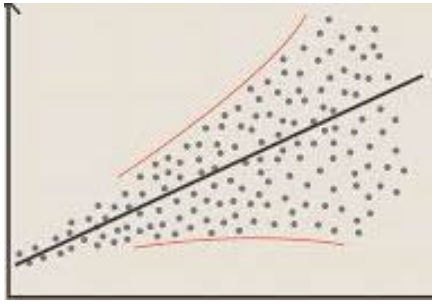
- **Linearity**: The relationship between X and the mean of Y is linear.
- **Normality**: For any fixed value of X, Y is normally distributed. We presume that deviations ϵi are picked from normal distribution. From this follows that the sampling distributions of slope and intercept (b1 & b0) are also normally distributed.
- **Independence**: Observations are independent of each other.
  We assume that our sample is a SRS (simple random sample) of size "n". This means that it consists of "n" points chosen from the population in such a way that every set of n individuals has an equal chance to be the sample actually selected.
- **Homoscedasticity**: The variance of residual is the same for any value of X.



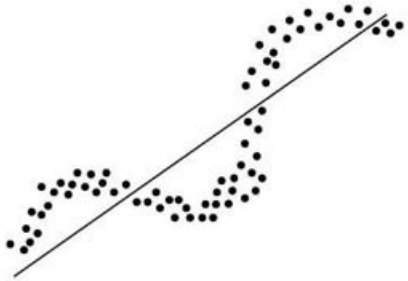
Linearity, normality,
same deviation (same σ) - **Homoscedasticity**

Note:
even if the ϵi are not Normally distributed, a general form of the central limit theorem tells us that the distributions of slope and intercept will be approximately Normal if number "n" is big enough.

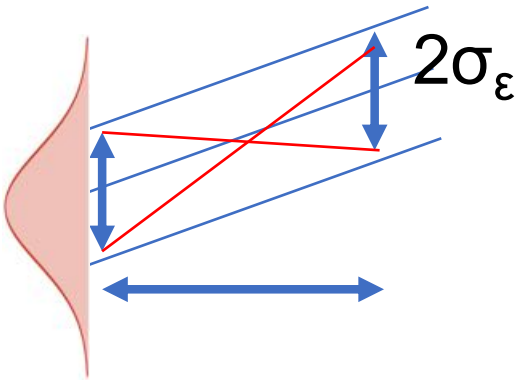
Heteroscedasticity
Variance changes



Autocorrelation.
Observations are NOT independent



Linear Regression – Slope Error



ϵ_i are Normally distributed with standard error σ. "x" is distributed uniformly ?
The error of the slope is calculated as:

$$\sqrt{\frac{\frac{1}{n-2}\sum_{i=1}^{n}\hat{\varepsilon}_i^{\,2}}{\sum_{i=1}^{n}(x_i-\bar{x})^2}} = \frac{\sigma_\varepsilon}{\sqrt{\sum(x-\bar{x})^2}}$$

Note:
even if the ϵ_i are not Normally distributed, a general form of the central limit theorem tells us that the distributions of slope and intercept will be approximately Normal if number "n" is big enough.

We do not know real population **σ**, we use the estimated model standard deviation **s**.
When we do this, we move from the Normal distribution to **t**-distributions with degrees of freedom **n – 2**, the degrees of freedom of **s**.

$$s = \sqrt{\frac{\sum e_i^2}{n-2}}$$

A confidence interval for the slope β1 is b1 ± t*SEb1
In this expression, t is the value for the t-distribution density curve with (n – 2) degrees of freedom with area between [−t, t].

To test the hypothesis H0: β1 = 0, compute the test statistic :

$$t = \frac{b_1}{\mathrm{SE}_{b_1}}$$

Example: 42 points, df= n-2 = 40.
Looking for a critical value in t-test table for df=40, P=0.05, two-tails combined: cv = 2.021

Then if slope b1=35, error SEb1=10,
then t = 35/10 = 3.5 >> 2.021, so we reject the null hypothesis that slope = 0

STANDARD DEVIATION OF THE REGRESSION SLOPE COEFFICIENT (POPULATION)

$$\sigma_{b_1} = \frac{\sigma_\varepsilon}{\sqrt{\sum(x-\bar{x})^2}}$$

where:

σ_{b_1}= Standard deviation of the regression slope (Called the standard error of the slope)

σ_ε= Population standard error of the estimate

ESTIMATOR FOR THE STANDARD DEVIATION OF THE REGRESSION SLOPE

$$s_{b_1} = \frac{s_\varepsilon}{\sqrt{\sum(x-\bar{x})^2}} = \frac{s_\varepsilon}{\sqrt{\sum x^2 - \frac{(\sum x)^2}{n}}}$$

where:

s_{b_1}= Estimate of the standard error of the least squares slope

$s_\varepsilon = \sqrt{\frac{SSE}{n-2}}$ Sample standard error of the estimate

Linear Regression – Slope Error via "r"

Standard Error of regression slope:

$$s(b_1) = \sqrt{\frac{1}{n-2} \cdot \frac{\sum (y_i - \hat{y}_i)^2}{\sum (x_i - \bar{x})^2}} \quad = \ldots = \sqrt{\frac{1}{n-2} \frac{sd(y)}{sd(x)} \sqrt{1 - r^2}}$$

where sd(...) is standard deviation, and **r – correlation coefficient.**

https://stats.stackexchange.com/questions/342632/how-to-understand-se-of-regression-slope-equation/342672

https://www.slideserve.com/karis/introduction-to-linear-regression-and-correlation-analysis

$$s(b_1) = \sqrt{\frac{1}{n-2} \cdot \frac{\sum (y_i - \bar{y})^2}{\sum (x_i - \bar{x})^2} \cdot \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}}$$

$$= \sqrt{\frac{1}{n-2}} \sqrt{\frac{\frac{1}{n-1} \sum (y_i - \bar{y})^2}{\frac{1}{n-1} \sum (x_i - \bar{x})^2}} \sqrt{\frac{SS_{\text{error}}}{SS_{\text{total}}}}$$

$$= \sqrt{\frac{1}{n-2}} \sqrt{\frac{sd(y)^2}{sd(x)^2}} \sqrt{\frac{SS_{\text{total}} - SS_{\text{model}}}{SS_{\text{total}}}}$$

$$= \sqrt{\frac{1}{n-2}} \frac{sd(y)}{sd(x)} \sqrt{1 - r^2}$$

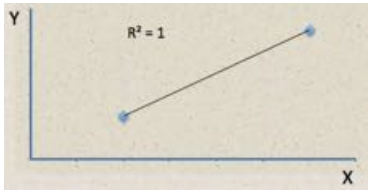
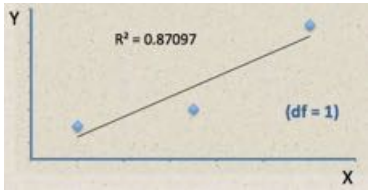
TEST STATISTIC FOR CORRELATION

$$t = \frac{r}{\sqrt{\frac{1 - r^2}{n-2}}}$$

$$df = n - 2$$

where:

t = Number of standard deviations r is from 0
r = Simple correlation coefficient
n = Sample size

df = degrees of freedom = n-2

If we have 2 points – we can draw only one line, no freedom, df = 0.
As we add points – we get more freedom.